

Introduction to MATH1860J

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SJTU Global College

AC Course Selection Workshop - 2026/9

Contents

1. About the Course
2. What to Learn & How to Learn
3. Student Feedback
4. Grading
5. Contact
6. Q & A Session

1. About the Course

Instructor Information: Prof. Horst Hohberger



Horst Harold Hohberger

Teaching Professor

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Education

Ph.D. (Dr. rer. nat.) University of Potsdam, Germany (2006)
Physics Diploma $\approx$ M.S. University of Potsdam (2001)
Chemistry pre-Diploma $\approx$ B.S. University of Potsdam (1995)

Work Experience

2021.09 – pres. Teaching Professor, UM-SJTU Joint Institute at the SJTU
2014.09 – 2021.08 Associate Teaching Professor, UM-SJTU Joint Institute at the SJTU
2007.03 – 2014.08 Lecturer, UM-SJTU Joint Institute at the SJTU
2001.10 – 2007.03 Research Assistant, University of Potsdam

Source: <https://gc.sjtu.edu.cn/about/faculty-staff/faculty-directory/>

Instructor Information: Prof. Luka Zwaan

Luka Zwaan / Personal webpage

Research

Teaching

Mentoring



Luka Zwaan

Postdoctoral researcher in
mathematics

📍 Belgium

🏛️ KU Leuven

✉️ Email

📄 arXiv

🟢 ORCID

Academic bio

- September 2024 - ongoing: Postdoctoral scholarship holder at KU Leuven
 - Department of Mathematics, Geometry Section
 - Working with [Marco Zambon](#)
- August 2018 - May 2024: PhD in Mathematics at University of Illinois at Urbana-Champaign
 - Advisor: [Rui Loja Fernandes](#)
 - Thesis: [Duistermaat-Heckman measures for Hamiltonian groupoid actions](#)
- September 2016 - August 2018: MSc. in Mathematical Sciences at Utrecht University
 - Master thesis: [Poisson Manifolds of Compact Types](#)
 - Supervised by [Marius Crainic](#)
- September 2013 - July 2016: BSc. in Mathematics at Utrecht University
 - Bachelor thesis: [The Conley-Zehnder Theorem](#)
 - Supervised by [Fabian Ziltener](#)
- September 2013 - July 2016: BSc. in Physics & Astronomy at Utrecht University

Source: <https://lukazwaan.github.io/>

Course Overview

MATH1860J is the first course in the *Honors Mathematics* sequence. It focuses on **single-variable functions**, with particular emphasis on:

- Mathematic Foundations
- Functions, Limits and Continuity
- Differentiation, Vector Spaces
- Series, Integration and Applications

Friendly Reminders:

- Lectures given by Prof. Hohberger will be online, while those given by Prof. Zwaan will be in-person. The content of two sections will be consistent and coherent.
- Compared to MATH1560J, MATH1860J is more theoretical and abstract. If you are interested in the theoretical induction and rigorous proof of mathematics, this course is for you!

Workload

The course workload is moderate to heavy. It requires consistent effort throughout the semester, since you might spend time contemplating the mathematical reasoning. Some assignment problems are also rather challenging to figure out. The workload includes:

- Lectures: $2.5 * 100$ min/week
- Recitation Classes: $\approx 1 * 120$ min/week, led by TAs, not mandatory but recommended.
- Assignments: One problem set per week, due the following week. (You will be assigned to an assignment group of 3 students. You are expected to work together with your group members to complete the assignments.)
- Midterm and Final Exams Preparation.

2. What to Learn & How to Learn

What to Learn: Course Outcomes

After completing MATH1860J, students should be able to:

Lecture	Lecture Subject
1	Elements of Logic
2	Set Theory
3	Natural, Rational and Real Numbers
4	Complex Numbers
5	Sequences and Convergence
6	Sequences and Convergence
7	Metric Spaces, Real Functions
8	Limits and Landau Symbols
9	Properties of Continuous Functions
10	Properties of Continuous Functions
11	Midterm Exam
12	The Derivative
13	The Derivative
14	Properties of Differentiable Functions
15	Properties of Differentiable Functions
16	Vector Spaces
17	Sequences of Functions
18	Series
19	Series
20	Power Series
21	The Exponential and Logarithm Functions
22	The Trigonometric Functions
23	Step Functions and Regulated Functions
24	The Regulated and Darboux Integrals
25	The Riemann Integral
26	Integration in Practice
27	Integration in Practice
28	Applications of Integration
29	Applications of Integration
30	Final Exam

- Compute limits of sequences and functions.
- Apply continuity and derivative theorems (IVT, MVT, convexity, extrema).
- Analyze convergence of series and power series.
- Evaluate integrals using various techniques.
- Construct and generalize all the outcomes rigorously.
- Experience an interesting journey to grasp advanced mathematics.

Source: [MATH186_course_description_2025](#)

How to Learn It Well I

Differences between Elementary Math and Advanced Math:

- Define concepts clearly $\rightarrow$ Certainty.
- Use what is certain to model real-world problems $\rightarrow$ Modeling.
- Generalize and prove with mathematical tools $\rightarrow$ Thinking.

“I’m good at math in high school” $\nRightarrow$ “I learn VV186/MATH1860J well.”

Please forget everything you have learned in school; you haven't learned it.

Please keep in mind everywhere the corresponding portions of your school work; you haven't actually forgotten them.

Edmund Landau, Foundations of Analysis, 1929

How to Learn It Well II

TAs' Suggestions:

- Attend lectures and follow the instructor's pace. Focus on key points and general ideas instead of memorizing every detail.
- Preview and review the lecture slides carefully, make sure you grasp every concept, theorem and definition, and understand the proof.
- Get familiar with mathematical analytical expression and notation. Don't be afraid of the abstractness of advanced mathematics.
- Do the assignments, quizzes and sample exams by yourself. Don't overrely on provided solutions or AI.
- Attending recitation classes (RCs) might help you to organize the knowledge framework.
- Actively discuss with your classmates. Post your questions on piazza.
- You could always ask for help from our kind TAs and instructors during office hours. Don't hesitate to ask questions!

3. Student Feedback

Comments on This Course I

*"Math186 is a really interesting course, where you obtain a lot while suffering from its abstract parts. However, enjoy your journey with 186 and remember to "bother" your TAs when suffering. **When you suffer, rush to canvas-files-sample_exams.**"*

"Math186 is more similar to "Mathematics analysis", which focuses more on how to proof and why the theorems work, instead of how to calculate the results. It is more abstract than 156 and harder to get started. But when you get familiar with those concepts and terms, you will find the beauty of math."

“一门需要抛弃高中知识从头开始学的课（hst语），但是186不像想象中‘纯证明’那么难，认真做作业和sample exam还是可以拿到不错的分数的。”

Comments on This Course II

“Perfect course, perfect professor. Your effort will certainly pay off as long as you really dive into the course and learn it by heart.”

“数分是一个很值得沉下心认真学习的课，只要愿意认认真真学习，不断积累沉淀，可以拿到一个不错的分数！”

“不知道我会不会是比较负面的回答。可能怪我自己学习方法和态度的问题。不只是186，很多JI课程我都是那种临时抱佛脚的状态。尽管最后成绩还算可以，但是对我来说，我其实并没有学到很多知识，基本都是考试前能对这门课的整个知识点有个大概的掌握，但是到了下学期，我对于这些知识会忘记很多。可能是我自己的学习问题，不过个人对于186的学习经历还挺负面的。”

4. Grading

Grading Policy

- ▶ The grade will be composed of the course work and the exams as follows:
 - ▶ Participation: 5 points
 - ▶ Course Outcome Quizzes: 5 points
 - ▶ Midterm exam: 45 points
 - ▶ Final exam: 45 points
- ▶ The actual grading scale will **usually** be based on the top approximately 6-12% of students receiving a grade of A+, with the following grades determined by (mostly) fixed point increments.
- ▶ Apart from this normalization, the grade distribution is up to you! If (for example) all students obtain many points in the exams, I am happy to see everyone receive a grade of A. Students are primarily evaluated with respect to a fixed point scale, not with respect to each other.

5. Contact

Contact

You could join the unofficial Q & A group chat if you have any questions or concerns about the course.

If you fail to join the group chat by QR code, please contact us via email, Feishu or WeChat.



**Group: 26FA-MATH1860J-
Q&A**



Valid until 9/5 and will update upon joining group

- Horst Hohberger, *math186_all_lecture_slides (2025)*
- MATH186_course_description_2025
- Wenxin Chen, *Introduction to Vv186*, AC Course Selection Workshop - 2025/9

6. Q & A Session

Thanks for Listening!

Any questions? Please feel free to ask!